



Shinrai

A Crowdsourced Civic Issue Reporting & Resolution System
From Concern to Completion

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The Problem We're Solving

Today, citizens have no easy, multilingual, or multimedia-based way to report local issues like potholes, garbage, and broken streetlights. On the governance side, complaints remain siloed across departments with no AI-based routing, no analytics, and no accountability loop. Adding to this, existing CCTV infrastructure remains entirely passive, capable of recording but unable to automatically detect accidents or trigger real-time alerts to authorities.

THE HINDU

A response by the Ministry of Consumer Affairs, Food and Public Distribution in Parliament shows that as of January 30, 2024, a total of 5.43 lakh consumer complaints were pending before district, State and national consumer commissions.

In 2024, the commissions received 1.73 lakh fresh cases but disposed of only 1.58 lakh, leading to a net increase of nearly 14,900 cases. The trend continued in 2025. Up to July this year, 78,031 new complaints were filed, while 65,537 cases were disposed of.

This backlog persists despite clear statutory timelines under the Consumer Protection Act, 2019. Section 38(7) of the Act mandates that complaints be decided within three months where no testing or analysis is required, and within five months where such examination is necessary.

Shinrai - Click, Report, Resolve.

Objective

We selected the Shinrai, a Crowdsourced Civic Issue Reporting System to address common civic problems such as potholes, broken roads, poor drainage, and waterlogging that frequently affect public safety and daily life. This project leverages AI technologies like NLP and Deep Learning along with a full-stack platform to allow citizens to report issues in real time. The system automatically categorizes complaints and forwards them to the relevant authorities for faster resolution. By enabling timely action, this solution can help reduce accidents, improve infrastructure, promote citizen participation, and support data-driven governance. The goal is to build a scalable system that solves real-world problems and positively impacts society at a large scale.



Project Overview

Enable Multi-Modal Civic Reporting —

Allow citizens to report issues via photo, video, voice, or text with GPS tagging, in multiple Indian languages

Automate Issue Classification & Routing —

Use AI (CNN/ANN) to categorise complaints by type and urgency, and route them to the correct department automatically

Provide Real-Time Tracking & Notifications —

Ensure every citizen gets live status updates via SMS, Email, or WhatsApp from submission to resolution

Empower Authorities with a Smart Dashboard —

Give government staff a unified portal to filter, assign, track, and close complaints with analytics and heatmaps

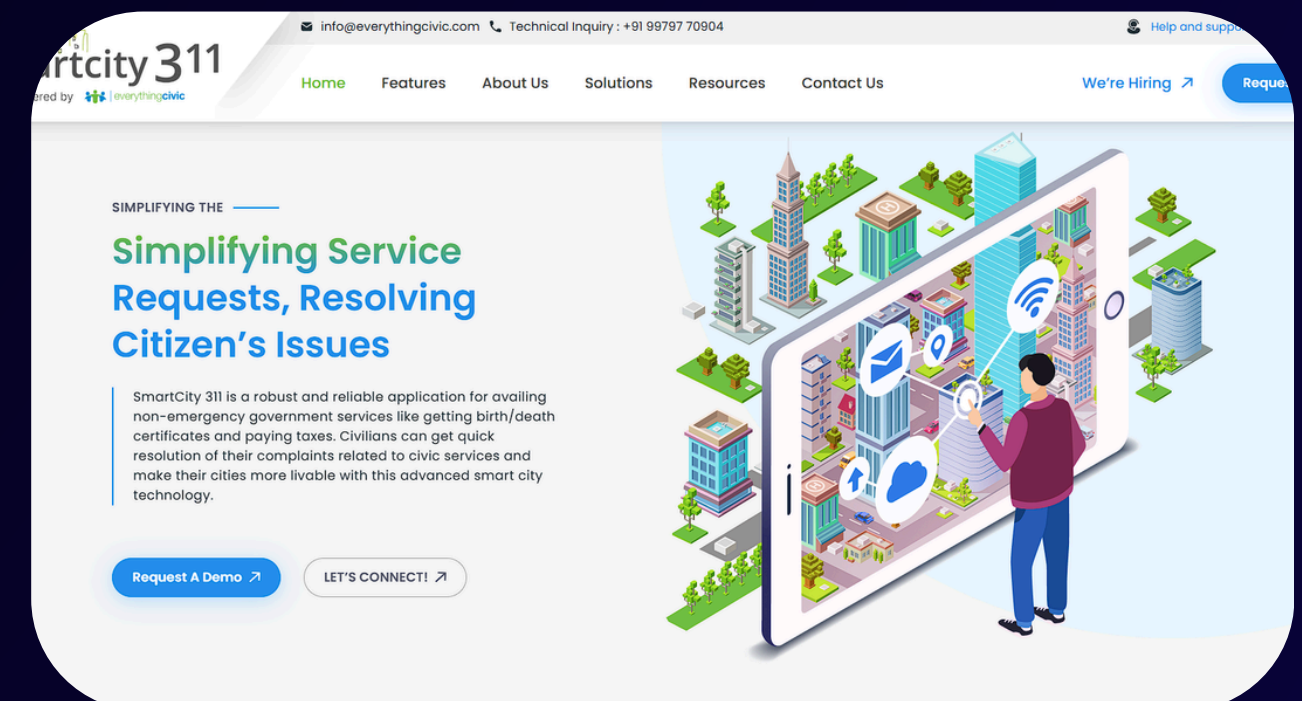
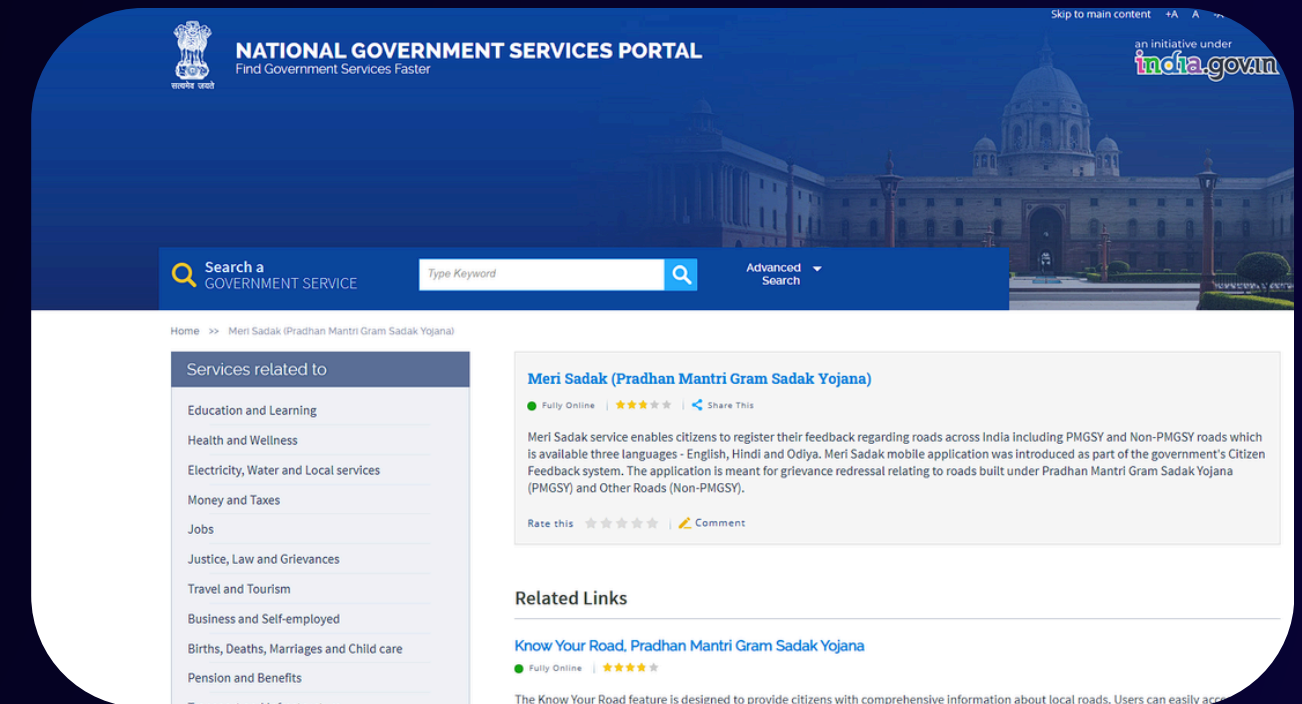
Detect Road Incidents via CCTV Integration —

Deploy YOLOv8-based real-time video analysis to auto-detect accidents and generate alerts + reports without human monitoring



Existing Solution

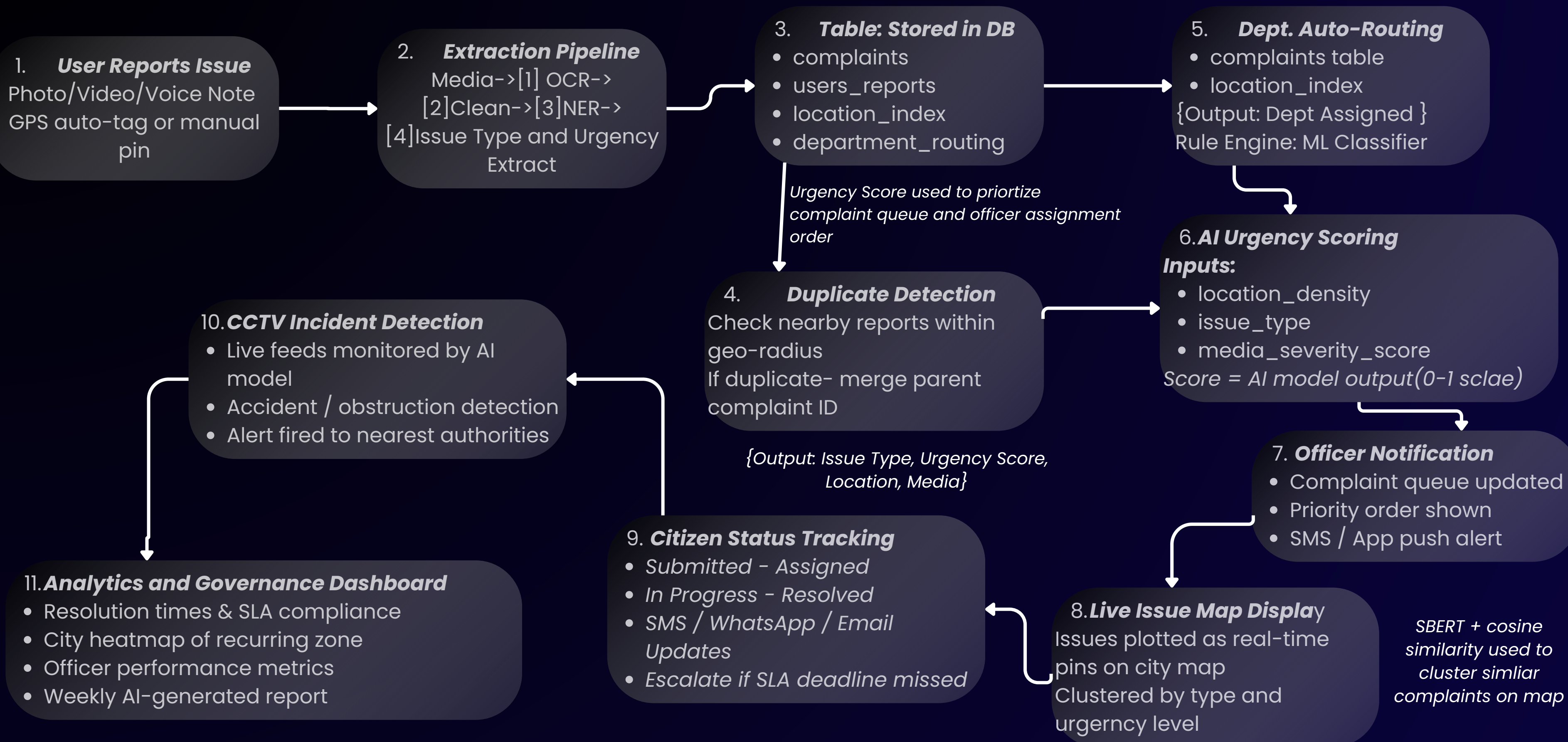
- Text & image only
- Keyword urgency buckets
- Manual department assignment
- Duplicate records created
- Self-reported closure
- Static admin dashboard
- No infrastructure monitoring dispatch
- No SLA enforcement
- Basic reports digest





System Architecture

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WHAT MAKES SHINRAI DIFFERENT

Existing tools accept only typed text or a single photo.

Shinrai accepts voice notes, video, OCR extracted text, and images all processed through a unified AI pipeline in multiple Indian languages.

Keyword matching labels urgency as High/Medium/Low. Our continuous AI urgency score (0–1) combines location density, media severity, and issue type, giving officers a precise, ranked queue, not a bucket.

Traditional systems create a new entry for every report. Shinrai uses SBERT semantic similarity + geo-radius matching to detect duplicates and merge them under a single parent complaint, with upvote count growing

Most systems let officers self-report resolution. Shinrai requires before and after image submission by field workers before a complaint can close, building a verified evidence trail for every issue.



DATASET DESCRIPTION

Civic Issue Dataset

This dataset contains images and complaint descriptions related to common civic problems such as:

- Potholes
- Garbage accumulation
- Water leakage
- Broken streetlights
- Damaged roads
- Traffic obstruction

CCTV Accident Dataset

Contains traffic surveillance videos and accident footage collected from public datasets and online surveillance sources.

Dataset Features

- Video frames
- Accident labels
- Vehicle movement data
- Collision events
- Time-based sequences

Text Complaint Data

Contains multilingual civic complaint texts collected from public complaint portals and synthetic datasets.

Dataset Features

- Complaint text
- Language labels
- Issue category
- Urgency keywords

Geo Location Dataset

Stores latitude and longitude information for mapping civic issues.

Dataset Features

- GPS coordinates
- Area name
- Complaint density
- Department mapping

Title: Civic Complaint Registration Application for Citizens of Urban/Rural Areas
Authors: Sandeep G. Shukla , Tejas A. Bhandare, Rohit V. Bava, Ishan S. Ahirrao
Year: 2024

Publisher: Journal of Emerging Technologies and Innovative Research |

Link: <https://www.jetir.org/papers/JETIR2405195.pdf>



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Technique used

- Image Processing & Computer Vision (Faster R-CNN)
- Machine Learning for Auto Classification
- Geo-tagging & Geo-fencing (GPS Integration)
- Automated Complaint Routing System
- Cloud-based Backend for Scalability
- User-friendly Mobile Interface

Limitation

- Dependence on Internet Connectivity
- Image Misclassification Possibility
- No Real-time Image Verification
- Limited Awareness & Adoption
- Data Privacy & Security Challenges
- Language/Accessibility Barriers

Advancement

- Image-based Complaint Reporting
- Fully Automated Workflow
- Faster Issue Resolution
- Real-time Status Tracking
- Increased Citizen Engagement
- AI-driven Smart Governance



Technique used

- Multimodal AI (Image + Text understanding)
- Large Language Models (MLLMs)
- Image-based civic issue detection
- Automated classification & prioritization
- AI-assisted decision support system

Limitation

- No real-world implementation or prototype
- No performance evaluation (accuracy, precision, etc.)
- High computational cost due to MLLMs
- No UI/UX or user adoption strategy
- Limited focus on data privacy and fake reporting
- Lacks complete workflow integration

Advancement

- Introduces multimodal AI approach for civic issue reporting
- Moves beyond traditional ML to vision + language understanding
- Proposes automated issue description generation from images
- Enables intelligent prioritization of civic issues
- Provides a scalable framework for smart city systems
- Reduces manual effort in issue validation and classification



Technique used

- Streamlit web app for submitting complaints with text, images, location, and contact details.
- Converts text into TF-IDF features for analysis.
- Uses Multinomial Naive Bayes to classify complaint types (Road, Water, Electricity, etc.).
- Detects urgency using keyword-based rules (e.g., "accident", "immediate").
- Stores complaint data (ID, category, urgency, location, image) in CSV/Excel.

Limitations

- Depends on user-written text → variations affect accuracy.
- Rule-based keywords for urgency & location → may miss complex cases.
- No clear accuracy metrics or large-scale testing → performance uncertain.
- No multilingual support → limited in diverse environments.
- Uses simple models (TF-IDF + Naive Bayes) → limited capability.
- CSV/Excel storage → scalability issues with large data.

Advancement

- Web-based Streamlit platform for submitting complaints (text, images, CSV).
- Uses NLP (TF-IDF + Naive Bayes) to classify complaints, detect urgency, and extract location.
- Real-time dashboards with charts and word clouds for monitoring trends.
- Fast processing and structured storage (CSV/Excel) for tracking.

Title : Smart Civic Issue Reporting System

Authors : Dnyanesh Walwadkar , Jayesh Patil , Mujahid Hussain , Saurav Yadav

Year : March 2022

Publisher : International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)

[Link](#)



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Technique used

- Android app with complaint submission (text, image, GPS).
- Three-tier MVC architecture for system organization.
- CNN–RNN model for image severity detection.
- SVM–based NLP for text classification and severity.
- Uses NNAPI for efficient on-device processing.
- Prioritizes complaints based on severity levels.

Limitations

- Depends on Android smartphones → limits accessibility for all users.
- No guarantee of response or resolution by authorities.
- Lacks detailed accuracy metrics and performance evaluation.
- Sensitive to poor-quality images or vague user input → affects accuracy.
- Higher computational demand of CNN–RNN may affect device performance and battery life.

Advancement

- Integrated smart-city system combining complaint submission, severity detection, and geo-based routing.
- Hybrid CNN–RNN model for prioritizing issues based on image severity.
- SVM–based NLP for analyzing text alongside images (multi-modal approach).
- Uses NNAPI for efficient on-device ML processing and faster performance.



Technique used

- Deep Learning-based approach
- Convolutional Neural Networks (CNN) for spatial feature extraction
- Recurrent Neural Networks (RNN) for temporal analysis
- Video-based accident detection using surveillance footage
- Automated classification of accident vs non-accident events
- Real-time monitoring system

Limitations

- Requires large labeled video datasets
- Performance affected by low light, weather, and occlusions
- High computational cost for real-time processing
- Limited testing in real-world deployment scenarios
- Possible false positives in crowded traffic scenes
- Dependency on camera quality and positioning

Advancement

- Provides real-time accident detection → faster emergency response
- Reduces human dependency in traffic monitoring
- Combines spatial + temporal learning (CNN + RNN)
- High accuracy (~98%) in detecting accidents
- Can be integrated with smart city infrastructure
- Improves road safety and reduces fatality rates

PROTOTYPE SCREENSHOT

Welcome Back

Sign in to your SHINRAI account

Select Your Role

Student
Report campus issues

Faculty/HOD
Verify and manage issues

Operational Staff
Execute tasks

Administrator
System monitoring

Email Address

you@university.edu

Password

••••••••

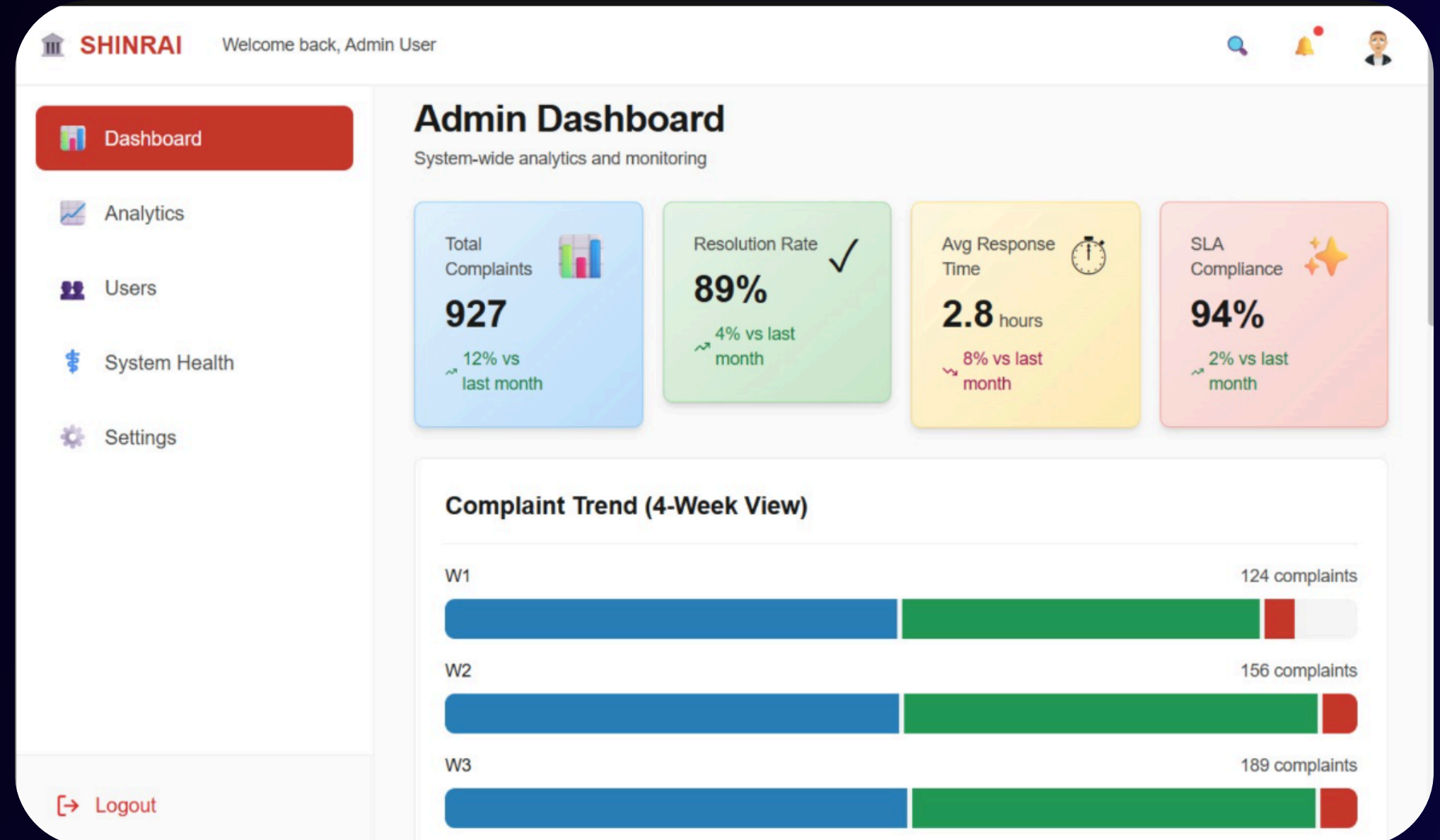
☐ Remember me

[Forgot password?](#)

Sign In

[Don't have an account? Sign up here](#)

Need help? [Contact Support](#)





Aligned SDGs

11 SUSTAINABLE CITIES
AND COMMUNITIES



Sustainable Cities and Communities:

This is your core SDG. Shinrai directly addresses urban civic issue reporting, better infrastructure management, and citizen participation in local governance.

16 PEACE, JUSTICE
AND STRONG
INSTITUTIONS



Peace, Justice and Strong Institutions: *Your authority portal, transparent complaint tracking, SLA accountability, and anti-corruption through public visibility directly support building accountable*

10 REDUCED
INEQUALITIES



Reduced Inequalities: *Multilingual support, voice input, anonymous reporting, and offline mode ensure that citizens from all literacy levels, languages.*



Project Timeline

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Phase	Task	Weeks
Phase 1 — Research & Design	Requirement gathering, wireframes, database schema design, API contracts	4-6
Phase 2 — Core App Development	Citizen mobile app (React Native), login/authentication, report submission flow	6-14
Phase 3 — AI Integration	CNN classifier training, ANN urgency scoring, RNN/LSTM timeline model, routing logic	14-18
Phase 4 — CCTV Module	YOLOv8 integration, alert pipeline, video clip + report generation	18-22
Phase 5 — Testing & Deployment	Unit & integration testing, Azure deployment, load testing, security audit	22-28



Team Structure

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Role	Task
Leadership Role	Harshita
Research	Purab, Piyush
Frontend/Design : React Native, AI tools	Rajat, Piyush
Backend: Nodejs	Rajat, Shreeshanth
AI Integration	Harshita, Purab, Poorna Charndra



Thank You